

DR. SARAH CHEN — PRINCIPAL AI SYSTEMS ARCHITECT

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Executive Summary

12+ years specializing in distributed systems, vector retrieval engines, and LLM inference optimization. Led architectures serving 500M+ daily embeddings with sub-15ms P99 retrieval latency.

Core Technical Competencies

Languages: Python, Rust, C++, Go, TypeScript. Infrastructure: Qdrant, Milvus, Kubernetes, Docker, Ray, Triton Inference Server, vLLM, TensorRT-LLM, CUDA 12.x.

Professional Experience

VP of AI Infrastructure — HyperScale Labs (2022–Present)

- Architected multi-region hybrid semantic search engine indexing 800M vectors across 64 nodes, achieving 99.995% uptime.
- Cut LLM token serving latency by 42% through speculative decoding and FlashAttention-3 integration.
- Reduced annual cloud expenditure by \$1.8M by designing an automated dynamic context pruning heuristic.

Lead Machine Learning Engineer — Cognitive Dynamics (2018–2022)

- Designed distributed document chunking and embedding pipelines processing 12TB of unstructured PDF datasets per day.
- Implemented hierarchical HNSW graph indexing algorithms reducing RAM footprint by 35% without recall degradation.
- Mentored an engineering group of 18 machine learning and data infrastructure engineers.

Education & Patents

- **Ph.D. in Computer Science**, Stanford University, 2018. Dissertation: 'Accelerated Approximate Nearest Neighbor Search in High-Dimensional Manifolds'.
- **B.S. in Electrical Engineering & CS**, UC Berkeley, 2014. Summa Cum Laude.
- **US Patent #11,489,203**: 'Adaptive Vector Quantization for High-Throughput Approximate Nearest Neighbor Lookups'.